



SAMANTHA LEE

AI Native | Pipeline Development | Computer Graphics

smlee.graphics@gmail.com · (484) 222-1335
San Francisco, CA · www.samanthamlee.com

EDUCATION

University of Pennsylvania

BSE in Computer Science (Digital Media Design)

SUMMARY

Senior Technical Artist building production tools and immersive experiences at Meta, Applied Intuition, and ILMxLAB. Specializes in pipeline development, Unreal Engine tooling, and real-time graphics, with hands-on experience integrating AI-based workflows to accelerate game and content development.

EXPERIENCE

Meta Reality Labs

Burlingame, CA

Senior Technical Artist

2026 - Present

- Developed AI-assisted game development workflows that empowered users to build games across platforms — investigated LLM-based 3D modeling, game design, and tool authoring skills
- Led a team to rapidly prototype different generative AI solutions for blocking out geometry to test gameplay, collaborating with game designers, artists, and engineers to optimize asset creation

Technical Artist

2025 - 2026

- Shipped switchable Home Vistas at Meta Connect 2025 — drove the fog transition tool with engineering, owned scene transition optimization, and reduced Quest 2 APK size by 30% via custom texture resolution settings
- Owned the lightprobe export tool powering dynamic avatar lighting across Horizon Home and Events; led content migration to Perforce and shipped a versioned toolchain that unblocked home environment delivery

Technical Artist (Contract)

2024 - 2025

- Built Unreal Engine C++ tooling for light baking, lightmap export, and asset processing — extending Meta's asset processor with custom logic to enable content ingestions to Meta Horizon Engine
- Drove tooling adoption with vendors and Meta artists, rapidly debugging asset processing, lightmap, and shader ingestion issues to unblock content delivery

Applied Intuition

Sunnyvale, CA

Technical Artist

2023 - 2024

- Architected custom C++ Unreal Engine pipelines to generate diverse photoreal environments for real-world autonomous driving simulations
- Designed and maintained a full-stack in-product material editing system that powered synthetic data generation pipelines for training autonomous driving AI models
- Developed sensor simulation and procedural world generation tools using HLSL shaders and Houdini, enabling industry-leading synthetic data quality

ILMxLAB

Remote

Technical Artist Intern

2022

- Created VFX elements including procedural vine growth shaders and environmental effects for VR and AR productions
- Maintained a scene capture and projection pipeline to optimize real-time rendering performance
- Collaborated with lighting and environment artists to achieve desired looks through custom Unreal tooling

Marvel Studios

Remote

Production Technology Intern

2021

- Built a web application to centralize and streamline footage requests across 20+ productions
- Improved dev/deploy pipeline using Unix/Linux tooling to increase codebase scalability
- Partnered with Disney Animation engineers to adapt a shared application for Marvel production environments

Apple

Remote

GPU Tools Intern

2021

- Contributed to low-level graphics developer tools including the GPU Shader Debugger
- Collaborated with UI/UX teams to design intuitive interfaces for graphics development tools
- Built tooling supporting GPU-accelerated machine learning, VR, and AR technologies

SKILLS

LANGUAGES & FRAMEWORKS

C/C++ · Python · Java · JavaScript · TypeScript · React Native · GLSL · OpenGL · USD · Unreal Blueprints

TOOLS

Unreal Engine · Houdini · Maya · Photoshop · Git · Perforce · RenderDoc · Claude Code